

MA 378 Summer Exam 20/21

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Q.1(a), First, we will show that if $p_n \in \mathcal{P}_n$ has $n+1$ zeros, then $p_n \equiv 0$

We know that $p_n(x_0) = p_n(x_1) = \dots = p_n(x_n) = 0$

This can be written as:

$$p_n(x) = (x-x_0)(x-x_1)\dots(x-x_n)q(x) \quad \text{for some remainder polynomial } q(x)$$

So the coefficient of x^{n+1} is $q(x)$. But $p_n \in \mathcal{P}_n$

$$\therefore q(x) = 0$$

$$\therefore p_n(x) \equiv 0 \quad \text{i.e. } p_n(x) = 0 \text{ for all } x \text{ if } p_n(x) \text{ has } n+1 \text{ zeros}$$

This can be used to show p_n is unique.

Suppose p_n is not unique, suppose that $p_n \in \mathcal{P}_n$ and $q_n \in \mathcal{P}_n$ both interpolate a function f at $n+1$ distinct points

$$p(x_k) = q(x_k) = y_k, \quad k=0, 1, \dots, n$$

$$\text{Let } r = p - q, \quad r \in \mathcal{P}_n$$

$$\text{Also } r(x_k) = p(x_k) - q(x_k) = 0 \quad \text{for } k=0, 1, \dots, n$$

Then r has $n+1$ zeros.

$$\therefore r \equiv 0$$

$$\therefore p = q, \quad p \text{ is unique.}$$

(b) Cauchy's Theorem: $f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \pi_{n+1}(x)$

$$|f(x) - p_n(x)| \leq \frac{M_{n+1}}{(n+1)!} |\pi_{n+1}(x)|$$

Cauchy's Theorem can not be used to conclude that $p_n(x) \rightarrow f(x)$ as $n \rightarrow \infty$.

This is true in some cases, such as when $a = -5$, $b = 5$ and $f(x) = e^{i\pi(x/4)}$

However there exists functions where $|f(x) - p_n(x)| \rightarrow \infty$ as $n \rightarrow \infty$, like Runge's example:

$$f(x) = \frac{1}{1+x^2} \text{ on } [-5, 5]$$

(c)

$$p_n(x) = \sum_{i=0}^n f(x_i) L_i(x)$$

$$L_i(x) = \prod_{j=0, j \neq i}^n \frac{x - x_j}{x_i - x_j}$$

$$f(x) = \sin\left(\frac{\pi x}{4}\right) \quad x_0 = 1, x_1 = 2, x_2 = 3$$

$$f(x_0) = \sin\left(\frac{\pi(1)}{4}\right) = \sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$f(x_1) = \sin\left(\frac{\pi(2)}{4}\right) = \sin\left(\frac{\pi}{2}\right) = 1$$

$$f(x_2) = \sin\left(\frac{3\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$L_0(x) = \left(\frac{x-2}{1-2}\right)\left(\frac{x-3}{1-3}\right) = \frac{x^2 - 5x + 6}{2}$$

$$L_1(x) = \left(\frac{x-1}{2-1}\right)\left(\frac{x-3}{2-3}\right) = \frac{x^2 - 4x + 3}{-1} = -x^2 + 4x - 3$$

$$L_2(x) = \left(\frac{x-1}{3-1}\right)\left(\frac{x-2}{3-2}\right) = \frac{x^2 - 3x + 2}{2}$$

$$\begin{aligned} p_2(x) &= \frac{1}{\sqrt{2}} \left(\frac{x^2 - 5x + 6}{2} \right) + 1(-x^2 + 4x - 3) + \frac{1}{\sqrt{2}} \left(\frac{x^2 - 3x + 2}{2} \right) \\ &= \frac{1}{2\sqrt{2}} (2x^2 - 8x + 8) - x^2 + 4x - 3 \\ &= \frac{1}{\sqrt{2}} (x^2 - 4x + 4) - x^2 + 4x - 3 \end{aligned}$$

$$\text{Check } p_2(1) = \frac{1}{\sqrt{2}} (1 - 4 + 4) - 1 + 4 - 3 = \frac{1}{\sqrt{2}} = \sin\left(\frac{\pi}{4}\right)$$

$$p_2(2) = \frac{1}{\sqrt{2}} (4 - 8 + 4) - 4 + 8 - 3 = 1 = \sin\left(\frac{\pi}{2}\right)$$

$$p_2(3) = \frac{1}{\sqrt{2}} (9 - 12 + 4) - 9 + 12 - 3 = \frac{1}{\sqrt{2}} = \sin\left(\frac{3\pi}{4}\right)$$

$$(d) |f(x) - p_2(x)| \leq \frac{M_3}{3!} |\pi_3(x)|$$

$$\leq \max_{1 \leq x \leq 3} |f'''(x)| |\pi_3(x)|$$

$$f(x) = \sin\left(\frac{\pi x}{4}\right)$$

$$f'(x) = \frac{1}{4} \pi \cos\left(\frac{\pi x}{4}\right)$$

$$f''(x) = -\frac{1}{16} \pi^2 \sin\left(\frac{\pi x}{4}\right)$$

$$f'''(x) = -\frac{1}{64} \pi^3 \cos\left(\frac{\pi x}{4}\right)$$

$$\begin{aligned} \max_{1 \leq x \leq 3} |f'''(x)| &= \max_{1 \leq x \leq 3} \frac{\pi^3}{64} \left| \cos\left(\frac{\pi x}{4}\right) \right| \text{ is @ } x=3 \text{ as } \cos\left(\frac{3\pi}{4}\right) = -\frac{1}{\sqrt{2}} \\ &= \left(\frac{\pi^3}{64} \right) \left(\frac{1}{\sqrt{2}} \right) \quad \text{and } -\frac{\pi^3}{64} \left(\frac{-1}{\sqrt{2}} \right) > 0 \\ &= \frac{\pi^3}{64\sqrt{2}} \end{aligned}$$

$$|f(x) - p_2(x)| \leq \frac{\pi^3}{64\sqrt{2}} \left(\frac{3}{8} \right)$$

$$|f(x) - p_2(x)| \leq \frac{3\pi^3}{256\sqrt{2}} = 0.2569$$

True error:

$$f(3/2) = \sin\left(\frac{\pi(3/2)}{4}\right) = \sin\left(\frac{3\pi}{8}\right) = 0.9239$$

$$\begin{aligned} p_2(3/2) &= \frac{1}{\sqrt{2}} \left(\left(\frac{3}{2}\right)^2 - 4\left(\frac{3}{2}\right) + 4 \right) - \left(\frac{3}{2}\right)^2 + 4\left(\frac{3}{2}\right) - 3 \\ &= \frac{1}{\sqrt{2}} \left(\frac{9}{4} - 6 + 4 \right) - \frac{9}{4} + 6 - 3 \\ &= \frac{1}{4\sqrt{2}} + \frac{3}{4} = 0.9268 \end{aligned}$$

$$\begin{aligned} \pi_3(x) &= (x-x_0)(x-x_1)(x-x_2) \\ &= (x-1)(x-2)(x-3) \\ &= (x^2-3x+2)(x-3) \\ &= x^3-3x^2+2x-3x^2+9x-6 \\ &= x^3-6x^2+11x-6 \end{aligned}$$

$$\begin{aligned} \pi_3(3/2) &= \left(\frac{3}{2}\right)^3 - 6\left(\frac{3}{2}\right)^2 + 11\left(\frac{3}{2}\right) - 6 \\ &= \frac{27}{8} - \frac{54}{4} + \frac{33}{2} - 6 \\ &= \frac{27-108+132-48}{8} \\ &= \frac{3}{8} \end{aligned}$$

$$\begin{aligned} &0.9239 \\ &- 0.9268 \\ \hline | - 0.0029 | &= 0.0029 \end{aligned}$$

True error of 0.0029 is much smaller than the upper bound of 0.2569.

$$2(a) \quad f(x_0) = \sin(0) = 0 \quad h = \frac{2-0}{2} = 1$$

$$f(x_1) = \sin\left(\frac{\pi}{2}\right) = 1$$

$$f(x_2) = \sin(\pi) = 0$$

For natural cubic spline we know:

$$\frac{1}{6} \sigma_{i-1} + 4\sigma_i + \sigma_{i+1} = \frac{1}{h^2} (f(x_{i-1}) - 2f(x_i) + f(x_{i+1})) \quad \text{and } \sigma_0 = \sigma_n = 0$$

With $i=1$ and $h=1$, we get

$$\frac{1}{6} (0 + 4\sigma_1 + \sigma_2) = (0 - 2f(x_1) + f(x_2))$$

$$\frac{1}{6} (4\sigma_1 + \sigma_2) = (-2 + 0)$$

$$4\sigma_1 + \sigma_2 = -12$$

$$\sigma_2 = \sigma_N = 0$$

$$4\sigma_1 = -12$$

$$\sigma_1 = -3$$

$$S_i(x) = \alpha_i (x - x_{i-1}) + \beta_i (x_i - x) + \frac{\sigma_{i-1}}{6h} (x_i - x)^3 + \frac{\sigma_i}{6h} (x - x_{i-1})^3, \text{ for } x \in [x_{i-1}, x_i]$$

$$\alpha_0 = \frac{f(x_0)}{h} - \frac{h}{6} \sigma_0 = 0 - 0 - \beta_1$$

$$\alpha_1 = \frac{f(x_1)}{h} - \frac{h}{6} \sigma_1 = 1 - \frac{1}{6}(-3) = 1 - \frac{1}{2} = \frac{1}{2} = \beta_2$$

$$\alpha_2 = \frac{f(x_2)}{h} - \frac{h}{6} \sigma_2 = 0 - \frac{1}{6}(0) = 0$$

$$S_1(x) = \alpha_0 (x - x_0) + \beta_1 (x_1 - x) + \frac{\sigma_0}{6h} (x_1 - x)^3 + \frac{\sigma_1}{6h} (x - x_0)^3 = \frac{1x}{2} - \frac{3}{6} (x - 0)^3 = \frac{1}{2} (x - x^3)$$

$x \in [x_0, x_1]$

$$S_2(x) = \alpha_2 (x - x_1) + \beta_2 (x_2 - x) + \frac{\sigma_1}{6h} (x_2 - x)^3 + \frac{\sigma_2}{6h} (x - x_1)^3 = \frac{1}{2} (2 - x) - \frac{1}{2} (2 - x)^3$$

$x \in [x_1, x_2]$

$$S(x) = \begin{cases} \frac{1}{2} (x - x^3) & x \in [0, 1] \\ \frac{1}{2} (2 - x) - \frac{1}{2} (2 - x)^3 & x \in [1, 2] \end{cases}$$

Check $s_1(1) = s_2(1)$

$$\frac{1}{2} (1 - 1) = \frac{1}{2} (2) - \frac{1}{2} (1)$$

$$0 = 0$$

$$\begin{aligned}
 (b) \quad S(1.5) &= \frac{1}{2}(2^{-3/2}) - \frac{1}{2}(2^{-3/2})^3 \\
 &= \frac{1}{2}\left(\frac{1}{2}\right) - \frac{1}{2}\left(\frac{1}{2}\right)^3 = \frac{1}{4} - \frac{1}{16} \\
 &= \frac{3}{16}
 \end{aligned}$$

$$|f(3/2) - S(3/2)| = \left| \sin\left(\frac{\pi(3/2)}{2}\right) - \frac{3}{16} \right|$$

$$\left| \sin\left(\frac{3\pi}{4}\right) - \frac{3}{16} \right|$$

$$\left| \frac{1}{\sqrt{2}} - \frac{3}{16} \right| = 0.5196$$

$$(c) \quad \frac{5h^4}{384} \max_{0 \leq x \leq 2} |f^{(4)}(x)| \leq 10^{-6}?$$

$$f(x) = \sin\left(\frac{\pi x}{4}\right)$$

$$f'(x) = \frac{1\pi}{4} \cos\left(\frac{\pi x}{4}\right)$$

$$f''(x) = -\frac{1\pi^2}{16} \sin\left(\frac{\pi x}{4}\right)$$

$$f'''(x) = -\frac{1}{64} \pi^3 \cos\left(\frac{\pi x}{4}\right)$$

$$f^{(4)}(x) = \frac{1}{256} \pi^4 \sin\left(\frac{\pi x}{4}\right)$$

$$\frac{5h^4}{384} \left(\frac{\pi^4}{256}\right) \leq 10^{-6}$$

$$\frac{5 \left(\frac{16}{N^4}\right) \left(\frac{\pi^4}{256}\right)}{384} \leq 10^{-6}$$

$$\frac{5}{24N^4} \left(\frac{\pi^4}{256}\right) \leq 10^{-6}$$

$$\frac{5(10^6)\pi^4}{(24)(256)} \leq N^4$$

$$\sqrt[4]{\frac{5(10^6)\pi^4}{(24)(256)}} \leq N$$

$$16.7795 \leq N$$

$$\max_{0 \leq x \leq 2} |f^{(4)}(x)| \text{ when } x=2: \frac{\pi^4}{256} (1)$$

N should take a value of 17 points to ensure that $\|f - S\|$ is no more than 10^{-6} .

(d) If $f \in C^4[a, b]$ and S is its cubic spline interpolant on $N+1$ equally spaced points then

$$\|f - S\|_{\infty} \leq \frac{5}{384} M_4 h^4$$

$$h = \frac{b-a}{N} \quad \text{and } M_4 \text{ is a constant}$$

$$\therefore h \rightarrow 0 \quad \text{as } N \rightarrow \infty$$

$$\therefore \frac{5M_4 h^4}{384} \rightarrow 0 \quad \text{as } N \rightarrow \infty$$

$$\therefore \lim_{N \rightarrow \infty} \|f - S\|_{\infty} = 0$$

and N can be increased to make our approximation as accurate as we want.

$$4. (b) i) u(x) = c_0 e^x + c_1 e^{-x} + \frac{x}{3}$$

$$u'(x) = c_0 e^x - c_1 e^{-x} + \frac{1}{3}$$

$$u''(x) = c_0 e^x + c_1 e^{-x}$$

$$-u''(x) + u(x) = \frac{x}{3}$$

$$-(c_0 e^x + c_1 e^{-x}) + c_0 e^x + c_1 e^{-x} + \frac{x}{3} = \frac{x}{3} \quad \dots \text{true}$$

$$(ii) u(0) = c_0 + c_1 = 0 \Rightarrow c_1 = -c_0$$

$$u(3) = c_0 e^3 + c_1 e^{-3} + 1 = 0$$

$$c_0 e^3 - c_0 e^{-3} = -1$$

$$c_0 (e^3 - e^{-3}) = -1$$

$$c_0 = \frac{-1}{e^3 - e^{-3}} \approx -0.0499$$

$$c_1 = \frac{1}{e^3 - e^{-3}} \approx 0.0499$$

b)

Multiply by $v(x)$ and integrate

$$-\int_0^3 u''(x) v(x) dx = \int_0^3 e^x v(x) dx$$

Integrate the first term by parts

$$-u'(x)v(x) \Big|_0^3 + \int_0^3 u'(x)v'(x) dx = \int_0^3 e^x v(x) dx$$

Since $v \in H_0^1(0,3)$ then $v(0) = v(3) = 0$, and the weak form is: find $u \in H_0^1(0,3)$ such that

$$\int_0^3 u'(x)v'(x) dx = \int_0^3 e^x v(x) dx$$

for all $v \in H_0^1(0,3)$

(C) The Galerkin basis functions are

$$w_i(x) = \begin{cases} (x-x_{i-1})/h & x_{i-1} \leq x < x_i \\ (x_{i+1}-x)/h & x_i \leq x < x_{i+1} \\ 0 & \text{otherwise} \end{cases}$$

The weak form is

$$\int_0^3 u(x) v'(x) dx = \int_0^3 e^x v(x) dx \\ = A(u, v) = L(v)$$

Let S be the space of piecewise linear functions on the uniform mesh $\{0, 1, 2, 3\}$ with $n=3$ intervals so $h = (3-0)/3 = 1$. Our numerical solution $u_h \in S$ is $u_h(x) = \lambda_1 w_1(x) + \lambda_2 w_2(x)$ so

$$A(u_h, v_h) := (u_h', v_h') = (e^x, v_h) \quad \text{for all } v_h(x) \in S$$

$$w_1(x) = \begin{cases} x & 0 < x < 1 \\ 2-x & 1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

$$w_2(x) = \begin{cases} 2-x & 1 < x < 2 \\ 3-x & 2 < x < 3 \\ 0 & \text{otherwise} \end{cases}$$

$$w_1'(x) = \begin{cases} 1 & 0 < x < 1 \\ -1 & 1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

$$w_2'(x) = \begin{cases} -1 & 1 < x < 2 \\ -1 & 2 < x < 3 \\ 0 & \text{otherwise} \end{cases}$$

$$(\lambda_1 w_1' + \lambda_2 w_2', w_1') = (e^x, w_1') \\ \lambda_1 \int_0^3 w_1' w_1' dx + \lambda_2 \int_0^3 w_2' w_1' dx = \int_0^3 e^x w_1' dx$$

$$(\lambda_1 w_1' + \lambda_2 w_2', w_2') = (e^x, w_2') \\ \lambda_1 \int_0^3 w_1' w_2' dx + \lambda_2 \int_0^3 w_2' w_2' dx = \int_0^3 e^x w_2' dx$$

$$\begin{aligned}\int_0^3 w_1' w_1' dx &= \int_0^1 w_1' w_1' dx + \int_1^2 w_1' w_1' dx + 0 \\ &= \int_0^1 1 dx + \int_1^2 (-1)^2 dx = \left. x \right|_0^1 + \left. x \right|_1^2 = 2\end{aligned}$$

$$\int_0^3 w_2' w_1' dx = 0 + \int_1^2 w_2' w_1' dx + 0 = \int_1^2 1 dx = \left. x \right|_1^2 = 1$$

$$\begin{aligned}\int_0^1 e^x w_2 dx &= \int_0^1 e^x x + \int_1^2 e^x (2-x) \\ &= \left. e^x + x e^x \right|_0^1 + \left. 2e^x - e^x - x e^x \right|_1^2 \\ &= 2e - 1 + 2e^2 - e^2 - 2e^2 - (2e - e - e) \\ &= 2e - 1 - e^2\end{aligned}$$

$$\int_0^3 w_1' w_2' dx = 0 + \int_1^2 w_1' w_2' dx + 0 = \int_1^2 1 dx = \left. x \right|_1^2 = 1$$

$$\begin{aligned}\int_0^3 w_2' w_2' dx &= 0 + \int_1^2 w_2' w_2' dx + \int_2^3 w_2' w_2' dx \\ &= \int_1^2 1 dx + \int_2^3 1 dx \\ &= \left. x \right|_1^2 + \left. x \right|_2^3 \\ &= 1 + 1 \\ &= 2\end{aligned}$$

$$\begin{aligned}\int_0^3 e^x w_2 dx &= \int_1^2 e^x (x-1) + \int_2^3 e^x (3-x) dx \\ &= \left. e^x + x e^x - e^x \right|_1^2 + \left. 3e^x - e^x - x e^x \right|_2^3 \\ &= e^2 + 2e^2 - e^2 - (e + e - e) + 3e^3 - e^3 - 3e^3 - (3e^2 - e^2 - 2e^2) \\ &= 2e^2 - e - e^3\end{aligned}$$

Then:

$$2\lambda_1 + \lambda_2 = -e^2 - 2e - 1 \approx -13.83$$

$$\lambda_1 + 2\lambda_2 = -e^3 + 2e^2 - e \approx -8.03$$

$$\lambda_2 = -13.83 - 2\lambda_1$$

$$\lambda_1 + 2(-13.83 - 2\lambda_1) = -8.03$$

$$\lambda_1 - 4\lambda_1 - 27.66 = -8.03$$

$$-3\lambda_1 = 18.63$$

$$\lambda_1 = -6.21$$

$$\begin{aligned}\lambda_2 &= -13.83 - 2(-6.21) \\ &= -1.41\end{aligned}$$

$$u_2(x) = \lambda_1 \varphi_1(x) + \lambda_2 \varphi_2(x)$$

$$u_2(x) = -6.21 \varphi_1(x) - 1.41 \varphi_2(x)$$